

Actions

2026-09-27

Definition 0.1 (Bekka and Mayer, 2000, ch. II, Section 2). An *action* of a group, Γ , on a measurable space $(X, \mathcal{B})$ is a measurable mapping

$$\Gamma \times X \rightarrow X, (\gamma, x) \mapsto T^\gamma x$$

with the following properties:

1. Associativity: For all $\gamma, \gamma' \in \Gamma, x \in X$, then $T^\gamma(T^{\gamma'}x) = T^{(\gamma\gamma')}x$.
2. Identity: There exists an identity element $e \in \Gamma$ such that $T^e x = x$ for all $x \in X$.
3. Quasi-Invariance: For any $B \in \mathcal{B}$ and for all $\gamma \in \Gamma$, we have $\mu(T^\gamma B) = 0$ if and only if $\mu(B) = 0$.

The action of Γ is also ergodic if it satisfies the additional property:

4. If $B \in \mathcal{B}$ and $\mu(B) = \mu(T^\gamma B)$ for any $\gamma \in \Gamma$, then $\mu(B) = 0$ or $\mu(X \setminus B) = 0$.

Remark 0.1. We don't require invertability in order to use actions and could instead use a monoid, M , defining the pre-image of M on $(X, \mathcal{B})$ as a measurable mapping

$$M \times X \rightarrow \mathcal{B}, (m, x) \mapsto (Tm)-1.x$$

such that

$$(Tm)-1.x = \{x' \in X : T^m x' = x\}.$$

Definition 0.2. A *topological dynamical system under the action of Γ* , denoted (X, Γ) , is a compact metric space X that has continuous surjective maps, $(\gamma, x) \mapsto T^\gamma x$, for all $\gamma \in \Gamma$.

Definition 0.3. Let $x \in X$, $\Phi = (\Phi_N)_{N \in \mathbb{N}}$ be a Følner sequence in Γ and μ a probability measure on X . Where δ_x is the Dirac mass at x , if

$$\frac{1}{|\Phi_N|} \sum_{\gamma \in \Phi_N} \delta_{T^\gamma x} \xrightarrow{\text{weakly}^*} \mu \text{ as } N \rightarrow \infty,$$

then we say x is generic for μ with respect to Φ and we denote this with $x \in \text{gen}(\mu, \Phi)$.¹

We are interested in how the action of a group Γ transforms functions on $(X, \mathcal{B}, \mu)$ so we must identify the associated definitions within functional analysis.

We define the map $U_\gamma : L^2(X) \rightarrow L^2(X)$ where $f \mapsto f \circ U_\gamma$ as the *Koopman operator* induced by $\gamma \in \Gamma$. We find that the Koopman operator induced by Γ is a unitary operator and the group homomorphism $U : \Gamma \rightarrow \mathcal{U}(L^2(X))$ is the *unitary representation* of Γ on $L^2(X)$, where $\mathcal{U}(L^2(X))$ is the set of all unitary operators on $L^2(X)$.

Bekka, M. B. and Mayer, M. (2000). *Ergodic theory and topological dynamics of group actions on homogeneous spaces*. Cambridge University Press.
<https://doi.org/10.1017/cbo9780511758898>.

¹consider tempered separately as the FCP construction only depends on sequential compactness